

Topic-wise Distribution

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Answer key is included after the MCQ section. The bonus MCQ is placed at the end of the PDF.

Trigonometry and Trigonometric Functions

Q1. What is the period of $f(x) = (1/2) \cos 2x$?

- a) 6π
- b) π
- c) $\pi/2$
- d) 4π
- e) 2π

Q2. What is the length of the arc of a circle of radius 1 cm and angle 120° ?

- a) 1.86 cm
- b) 2.09 cm
- c) 22.9 cm
- d) 1.5 cm

Q3. Find the length of the arc whose radius is 9 cm and central angle is 360° .

- a) 56.55 cm
- b) 69.16 cm
- c) 3140 cm
- d) 65.54 cm
- e) 74.13 cm

Q4. $\sin 30^\circ + \sin 16^\circ = ?$

- a) $2 \sin 14^\circ \cos 14^\circ$
- b) $-2 \sin 28^\circ \cos 14^\circ$
- c) $2 \sin 7^\circ \cos 23^\circ$
- d) $2 \cos 17^\circ \sin 14^\circ$
- e) $2 \sin 23^\circ \cos 7^\circ$

Q5. If $\cos(30^\circ - \alpha) - \sin(\alpha - 30^\circ) = 0$, where $0^\circ \leq \alpha \leq 90^\circ$, then $\alpha =$ ____.

- a) 15°
- b) 45°
- c) 75°
- d) 30°
- e) 60°

Q6. If $180^\circ \leq \theta \leq 360^\circ$ and $\theta = \tan^{-1}(-\sqrt{3})$, then $\theta =$ ____.

- a) 210°
- b) 330°
- c) 300°
- d) 240°
- e) 60°

Q7. $\sin 300^\circ$ equals ____.

- a) $\sin 120^\circ$
- b) $\cos 120^\circ$
- c) $\sin 240^\circ$
- d) $\cos 240^\circ$
- e) $\cos 30^\circ$

Q8. Which of the following is equal to $\tan \theta + \cot \theta$?

- a) $(\sec \theta)(\operatorname{cosec} \theta)$
- b) $2/\sec^2 \theta$
- c) $1/\cot \theta + \tan \theta$
- d) $(\sin \theta + \cos \theta)/(\sin \theta + \cos \theta)$
- e) $\tan^2 \theta + 1$

Q9. If $0 \leq x \leq 360^\circ$ and $4 \sin^2 x + 4 \cos x - 1 = 0$, which set contains all values of x ?

- a) $\{60^\circ, -60^\circ\}$
- b) $\{30^\circ, 60^\circ\}$
- c) $\{60^\circ, 120^\circ\}$
- d) $\{120^\circ, 240^\circ\}$
- e) $\{120^\circ, -120^\circ\}$

Q10. If $\cos x - 1 = -2 \sin((x+y)/2) \sin((x-y)/2)$, where $x \geq 0$ and $y \leq 90^\circ$, then $y =$ ____.

- a) 45°
- b) 30°
- c) 90°
- d) 60°
- e) 0°

Q11. In any triangle ABC, $a = 10$, $c = 3$, $\alpha = 90^\circ$, find $\sin \gamma$.

- a) 0.2
- b) 0.9784
- c) 1.2745
- d) 0.8485
- e) None of these

Q12. The slope of the tangent to the curve $y = e^{(\ln(\sin x))}$ at $x = 60^\circ$ is ____.

- a) 0.5
- b) 0.866
- c) 10
- d) $e^{(\ln 0.5)}$
- e) $e^{(\ln 0.866)}$

Q13. If $y = f(x) = -(5/2) \sin(-\sqrt{3} x + 7)$, then the range of $f(x)$ is ____.

- a) $5/2 \leq y \leq -5/2$
- b) $-1 \leq y \leq 1$
- c) $-\infty < x < \infty$
- d) $-\sqrt{3} \leq y \leq \sqrt{3}$
- e) $-5/2 \leq y \leq 5/2$

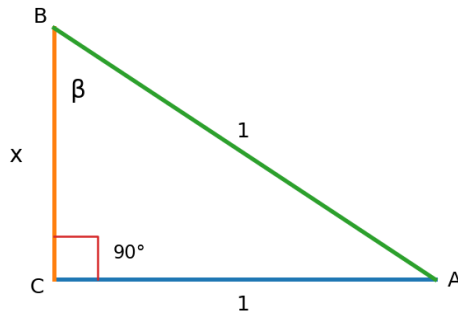
Q14. If $y = f(x) = -\sqrt{3} \tan 2x$, then the range of $f(x)$ is ____.

- a) $-2 \leq y \leq 2$
- b) $-60^\circ \leq y \leq 60^\circ$
- c) $-\infty < y < \infty$
- d) $-\sqrt{3} \leq y \leq \sqrt{3}$
- e) $\mathbb{R} - \{x \mid x = (2n+1)\pi/4, n \in \mathbb{Z}\}$

Q15. If $f(x) = e^{(\cos x)}$ on the interval $0 < x < \pi/6$, then $f(x)$ is ____.

- a) Neither increasing nor decreasing
- b) Information is incomplete
- c) Increasing as well as decreasing
- d) Decreasing
- e) Increasing

Q16. In the given figure, if $\sec \beta = \infty$, then $x =$ ____.



- a) $\pi/2$
- b) 0
- c) $\sqrt{2}$
- d) $\pi/4$
- e) 1

Differential Calculus

Q17. If $y = 5^x$, then $dy/dx = ?$

- a) 4
- b) -14
- c) 1
- d) $5^x \ln 5$
- e) $t(5^x \ln 5)$

Q18. Let $y = \cos x + \cos 2x$, then $dy/dx = ?$

- a) $2 \cos x(-\sin x)$
- b) $(1/2) \cos 2x$
- c) $-\sin x - 2 \sin 2x$
- d) $\sin x(2 \sin 2x)$

Q19. If $y = 1/\ln(1/x)$, then $d^2y/dx^2 =$ ____.

- a) $-\{ 1/[x^2(\ln x)^2] + 2/[x^2(\ln x)^3] \}$
- b) $-1/x^2$
- c) $-1/x^3$
- d) $1/[(\ln x)^2 x]$
- e) $-\{ 1/(x \ln x) + 2/[x^2(\ln x)^2] \}$

Q20. $\frac{d}{dx} \{ [(g(x))^m]^m \} = ?$

- a) $m[g(x)]^{(m-1)}$
- b) $m[(g(x))^m]^{(m-1)} [m(g(x))^{(m-1)} \frac{d}{dx}(g(x))]$
- c) $m[g(m-1)x]g'(x)$
- d) $m[[g(x)]^{(m-1)}]^{(m-1)}$
- e) $n[g(m-1)^{(n-1)}]g'(x)$

Q21. If $y = (\sin(1/x))^{-1}$, then $y' = ?$

- a) $\cos(1/x)/\{x \sin(1/x)\}^2$
- b) $\cos(1/x)$
- c) $(-1)(\sin(1/x))^{-1}(\cos(1/x))$
- d) $\cos(1/x)/(\sin(1/x))^2$
- e) $\cot(1/x)$

Q22. If $y = x e^{(ix)}$, where $i = \sqrt{-1}$, then $y'' = \underline{\hspace{2cm}}$.

- a) $2e^{(ix)}$
- b) $(2i - x)e^{(ix)}$
- c) $-e^{(ix)}$
- d) $(i + ix - x) - e^{(ix)}$
- e) $-x e^{(ix)}$

Q23. $\frac{d}{dx} [f(x^2)g(x)] = ?$

- a) $f(x^2)g(x) + f(x^2)g(x)$
- b) $f'(x^2)g'(x)$
- c) $2xf'(x^2)g(x) + f(x^2)g'(x)$
- d) $2xf'(x)g(x) + f(x)g'(x)$
- e) $f(x^2)g(x)$

Q24. If $y = \cos^{-1}(x^{-1})$, then $y' = ?$

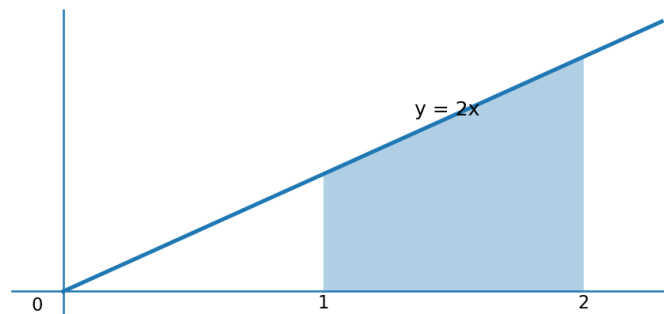
- a) $1/(-\sin x)$
- b) $\sec x \tan x$
- c) $-1/\sqrt{1 - x^2}$
- d) $1/\sqrt{1 - x^2}$
- e) $1/[x\sqrt{(x^2 - 1)}]$

Integral Calculus

Q25. $\int \sqrt{1 - \sin x} \, dx = \underline{\hspace{2cm}}$.

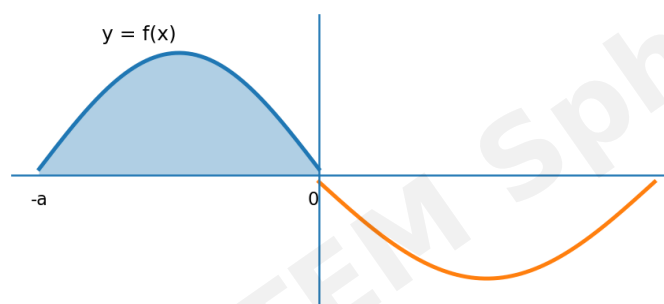
- a) $-2(\sin(x/2) + \cos(x/2)) + C$
- b) $\sqrt{1 + \cos x} + C$
- c) $\sqrt{x + \cos x} + C$
- d) None of these
- e) $\sin x - \cos x + C$

Q26. The area of the shaded portion in the figure is ____.



- a) 3
- b) $1/2$
- c) $1/9$
- d) $5x$
- e) $2x$

Q27. The area of the shaded region in the given figure is equal to ____.



- a) $2 \int_0^a f(x) dx$
- b) $(1/2) \int_0^a f(x) dx$
- c) $-\int_{-a}^0 f(x) dx$
- d) $\int_{-a}^0 f(x) dx$
- e) $2 \int f(x) dx$

Q28. $\int \sec^2 x dx = ?$

- a) $\tan x$
- b) $\sin x$
- c) $\cot x$
- d) $\sec x$
- e) $\operatorname{cosec} x$

Q29. The value of $\int_0^2 4e^{(2t)} dt$ is:

- a) $8e^4$
- b) $2e^4$
- c) $2(e^4 - 1)$
- d) e^4
- e) $2(e^4 + 1)$

Q30. $\int (\sin x - \cos x)^2 dx = ?$

- a) $x + (\sin 2x)/2$
- b) $x + (\cos 2x)/2 + C$
- c) $x - (\tan 2x)/2$
- d) $x - (\sec 2x)/2$
- e) $x - (\cos 2x)/2 + C$

Limits

Q31. $\lim_{x \rightarrow \infty} (1 + 3/x)^x = ?$

- a) 4
- b) e^3
- c) e^5
- d) e^2

Q32. $\lim_{x \rightarrow 0} (1 - a^{-x})(x a^{-x})^{-1} = \underline{\hspace{2cm}}$.

- a) ∞
- b) 0
- c) $\log_a e$
- d) 1
- e) $\log_e a$

Q33. $\lim_{x \rightarrow \infty} (x^3 - 3x^2 + 2x + 1)/(3x^3 - 4x^2 + 2x + 5) = ?$

- a) $1/3$
- b) $1/9$
- c) $-2/9$
- d) $-2/3$

Q34. $\lim_{x \rightarrow 0} (\sin^2 x - 1)/(\sin^2 x - 2\sin x + \cos^2 x) = \underline{\hspace{2cm}}$.

- a) $\sin x + 1$
- b) 1
- c) $\sin x - 1$
- d) -1
- e) 0

Sequences, Series and Binomial Theorem

Q35. If $S_n = \sum_{k=0}^n ar^k$ and $0 < r < 1$, then $S_\infty = ?$

- a) $a/(1-r)$
- b) $(1-r)/a$
- c) $1/(1-r)$
- d) $(1+r)/a$
- e) $a/(1+r)$

Q36. Find four numbers in A.P. whose sum is 24 and sum of squares is 164.

- a) 3, 6, 8, 7
- b) 3, 5, 7, 9
- c) 4, 3, 2, 9
- d) -3, 5, -6, 4
- e) -4, -3, -2, -1

Q37. If ${}^nC_5 = {}^nC_3$, then $n =$ ____.

- a) 2
- b) 8
- c) 10
- d) 5
- e) 15

Q38. If $y = \frac{1}{3} + \frac{(1 \cdot 3/2!)(1/3)^2}{1} + \frac{(1 \cdot 3 \cdot 5/3!)(1/3)^3}{1} + \dots$, then $y^2 + 2y =$ ____.

- a) 0
- b) 1
- c) 2
- d) 3

Q39. If the H.M. between 3 and x is 3, then $x =$ ____.

- a) $\frac{1}{3}$
- b) $\frac{9}{2}$
- c) $\frac{2}{3}$
- d) 0
- e) 3

Q40. $1 + 2 + 3 + \dots + n =$ ____.

- a) $2n + 1$
- b) $\frac{n(n+1)(2n+1)}{6}$
- c) $\frac{n(n+1)}{2}$
- d) $\left[\frac{n(n+1)}{6}\right]^2$
- e) n^2

Q41. Which term in the expansion of $(x/2 + 2/x^2)^{12}$ is independent of x ?

- a) 6
- b) 5
- c) 13
- d) 1

Algebra and Complex Numbers

Q42. The sum of five different positive odd integers each less than 16 is equal to the square of an integer x . The least value of x is ____.

- a) 6
- b) 5
- c) 7
- d) 8
- e) 3

Q43. The positive values of x for which $9x^{(2/3)} + 4x^{(-2/3)} = 37$ are:

- a) $(\frac{1}{27}, 8)$
- b) $(27, 8)$
- c) $(64, 27)$
- d) $(1, 125)$
- e) $(81, 9)$

Q44. The multiplicative inverse of $1 - 2i$ is _____, where $i = \sqrt{-1}$.

- a) $-1 + 2i$
- b) $2/5 + (1/5)i$
- c) $1/5 + (2/5)i$
- d) $3/5 + i$
- e) $1/4 + 5i$

Q45. If one root of $x^3 + 2x^2 + x + 2$ is i , where $i = \sqrt{-1}$, which set contains all other solutions?

- a) $\{-i, -2\}$
- b) $\{2\}$
- c) $\{-i\}$
- d) $\{-i, 2\}$
- e) $\{2, -2\}$

Q46. The imaginary term of $-i^2(2 - i)$ is _____.

- a) $2i$
- b) 2
- c) $2i^2$
- d) $-i$

Q47. If one root of $3x^2 - 6hx + h^2 - 25 = 0$ is zero, then the value of h is _____.

- a) 4
- b) 5
- c) -4
- d) -5
- e) ± 5

Coordinate Geometry and Conic Sections

Q48. The line $y = mx + c$ touches the parabola $y^2 = 4ax$ if _____.

- a) $m = a^2c$
- b) $c = m/a$
- c) $c = a/m$
- d) $c = 2a/m$
- e) $c = a\sqrt{1 + m^2}$

Q49. A conic is the set of points such that the distance of each from a fixed point bears a constant ratio greater than one to its distance from a fixed straight line. This conic is _____.

- a) Ellipse
- b) Directrix
- c) Parabola
- d) Circle
- e) Hyperbola

Q50. The point $P(0,b)$ lies inside the circle $x^2 + y^2 - 2 = 0$ if $b =$ ____.

- a) $-\sqrt{2}$
- b) 2
- c) -2
- d) $\sqrt{2}$
- e) -1

Q51. Two lines $ax^2 + 2hxy + by^2 = 0$ will be imaginary if ____.

- a) $h^2 < ab$
- b) $h^2 = ab$
- c) $ab = 0$
- d) $h^2 > ab$
- e) $h^2 = -ab$

Q52. A line L is perpendicular to the y -axis and the coordinates of a point P on L are $(-5,-3)$. What is α if $Q(5,\alpha)$ also lies on L ?

- a) -3
- b) 0
- c) 5
- d) 3
- e) -5

Q53. Two tangents drawn from a point P to the circle $x^2 + y^2 = 121$ may meet at a point other than P which lies at ____.

- a) No such point exists
- b) $x = 11$
- c) $x = \infty$
- d) $y = 11$
- e) $y = \infty$

Q54. If $f(x) = \sqrt{(x - 1)/x}$, what is the domain of $f(x)$?

- a) All real numbers less than or equal to -1
- b) All real numbers except 0
- c) All real numbers greater than or equal to 1
- d) All real numbers less than or equal to 1
- e) All real numbers greater than or equal to -1 but less than or equal to 1

Q55. The eccentricity of the hyperbola $x^2/25 - y^2/36 = 1$ is:

- a) $5/\sqrt{61}$
- b) $\sqrt{61}/5$
- c) $\sqrt{61}/9$
- d) $-\sqrt{61}/5$
- e) $-9/\sqrt{61}$

Q56. If $x^2/81 + y^2/121 = 1$ for an ellipse, then the length of latus rectum is:

- a) $172/11$
- b) $162/11$
- c) $163/21$
- d) $165/21$
- e) $242/81$

Q57. What is the equation of the line passing through the origin and making an angle of 30° anticlockwise with the line $x = 0$?

- a) $y = \sqrt{3}x$
- b) $y = -\sqrt{3}x$
- c) $2y = -\sqrt{3}x$
- d) $x = -\sqrt{3}y$
- e) $2y = \sqrt{3}x$

Vectors

Q58. $[a \ b \ b] = \underline{\hspace{1cm}}$.

- a) 0
- b) -1
- c) k
- d) i
- e) 1

Q59. Two or more vectors are said to be coplanar if they are parallel to the same $\underline{\hspace{1cm}}$.

- a) Vector
- b) Line
- c) Magnitude
- d) Plane
- e) Scalar

Q60. For what value of a will the vector $[1, 2a, -4]$ be perpendicular to the vector $[-3, 1, 2]$?

- a) $11/2$
- b) -3
- c) 4
- d) $2/11$
- e) -1

Q61. $i \times j = \underline{\hspace{1cm}}$.

- a) $-j \times i$
- b) $k \times i$
- c) zero
- d) $j \times i$
- e) $k \times j$

Matrices and Determinants

Q62. If A is a matrix such that $\det(A) = 3$ and B is obtained by interchanging the 1st and 3rd rows of A, then $\det(B) = \underline{\hspace{1cm}}$.

- a) $1/3$
- b) 6
- c) -3
- d) 3

Q63. Evaluate the determinant:

$$\begin{vmatrix} y^2z & z^2x & x^2y \\ x/y & y/z & z/x \\ y/x & z/y & x/z \end{vmatrix}$$

- a) 1
- b) 0
- c) xyz
- d) $x^2y^2z^2$
- e) -1

Q64. The value of the determinant is:

$$\begin{vmatrix} x^2-y^2 & y^2-z^2 & z^2-x^2 \\ y^2-z^2 & z^2-x^2 & x^2-y^2 \\ z^2-x^2 & x^2-y^2 & y^2-z^2 \end{vmatrix}$$

- a) -1
- b) 0
- c) 1
- d) $(x-y)(y-z)(z-x)$
- e) $(x^2-y^2)(y^2-z^2)(z^2-x^2)$

Sets, Binary Operations and Group Theory

Q65. If $A \subset B$ and $B \subset C$, then $A \cap C =$ ____.

- a) A'
- b) C
- c) ϕ (Null set)
- d) B
- e) A

Q66. If $A = \mathbb{N}$, the set of natural numbers, then which operation is not a binary operation on A?

- a) Division
- b) Multiplication and addition
- c) Multiplication
- d) None of these
- e) Addition

Q67. Let H be a non-empty set and * a binary operation in H. An element e of H is an identity element with respect to * if for all $a \in H$, ____.

- a) $a * e = 1$
- b) $a * e = e$
- c) $a * e = a^3$
- d) $a * e = a$
- e) $a * e = a^2$

Q68. In a group, how many inverse elements correspond to each element?

- a) More than one
- b) None
- c) Two
- d) Only one

Permutation, Combination and Probability

Q69. How many words can be formed from the letters of the word "DAUGHTER" if vowels are to be placed at second, fourth and sixth places?

- a) 740
- b) None of them
- c) 720
- d) 865

Q70. How many words can be formed out of the letters of the word "ASSOCIATION" when all the letters are used in each word?

- a) $11!/(3!3!3!)$
- b) $11!/(2!2!2!2!)$
- c) None of these
- d) $11!/(4!4!)$
- e) $11!/(2!2!2!)$

Q71. Two dice are thrown simultaneously. Event A: the sum of dots is greater than 6. Event B: the sum of dots is even. $P(A \cap B)$ equals:

- a) $5/6$
- b) $1/2$
- c) $9/36$
- d) $21/36$

Geometry of Triangles and Circles

Q72. If all three sides of a triangle are tangent to a circle, then the circle is called ____.

- a) Escribed circle
- b) Unit circle
- c) Circumcircle
- d) Outcircle
- e) In-circle

Q73. The incircle of a triangle touches all the sides at their midpoints. The triangle is ____.

- a) Right angled
- b) Isosceles
- c) Obtuse angled
- d) Equilateral

Q74. In triangle ABC, sides are $a = 3$, $b = 4$, $c = 5$. Then the first exradius $r_1 =$ ____.

- a) $5/2$
- b) $5/4$
- c) 2
- d) 1
- e) 3

Q75. If triangle ABC has sides 6, 8 and 10, then the circumference of its circumcircle is _____.

- a) 100π
- b) $10\pi^2$
- c) 10π
- d) 20π
- e) 10

Fakhar STEM Sphere

Bonus MCQ

Bonus Q. If $\vec{v} = a\hat{i} + a\hat{j} + a\hat{k}$ is a unit vector, then $a = \underline{\hspace{1cm}}$.

- a) $1/3$
- b) 0
- c) 1
- d) -1
- e) $1/\sqrt{3}$

Correct Answer: E) $1/\sqrt{3}$

Since $|\vec{v}| = \sqrt{a^2 + a^2 + a^2} = \sqrt{3a^2}$. For a unit vector, $\sqrt{3a^2} = 1$, so $a = 1/\sqrt{3}$.

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